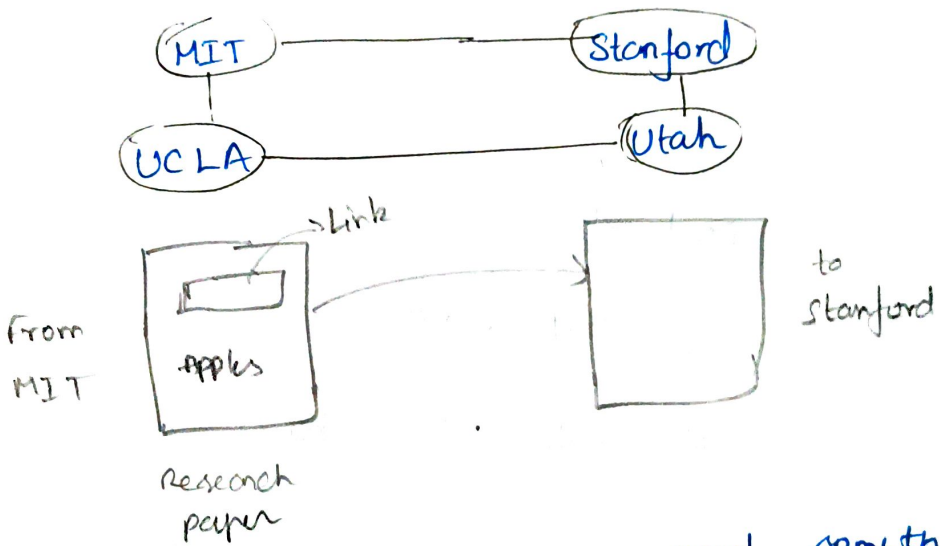


Computer Networking

Internet : Collection of computer network.
(Network of Networks)

ARPA built ARPANET
(Advanced Research Project Agency)

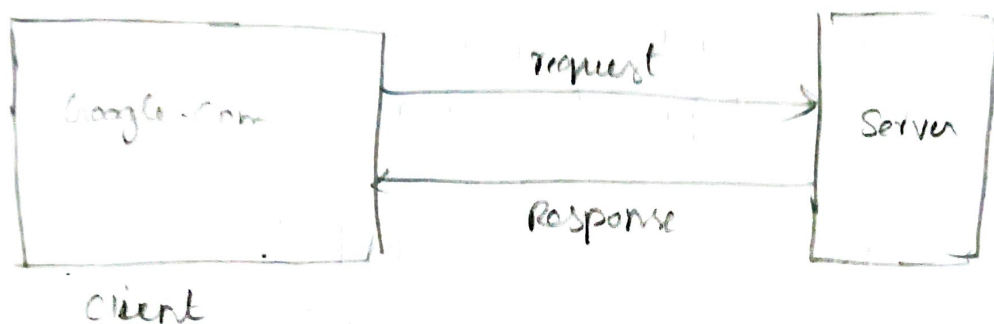


to open that link we need something
that is world wide web (WWW) commonly
known as web, is an information system
where documents & other resources are identi-
fied by Uniform Resource Locators (URLs)

Tim Berners-Lee - WWW

Protocols : set of rules how the files are shared across the network.

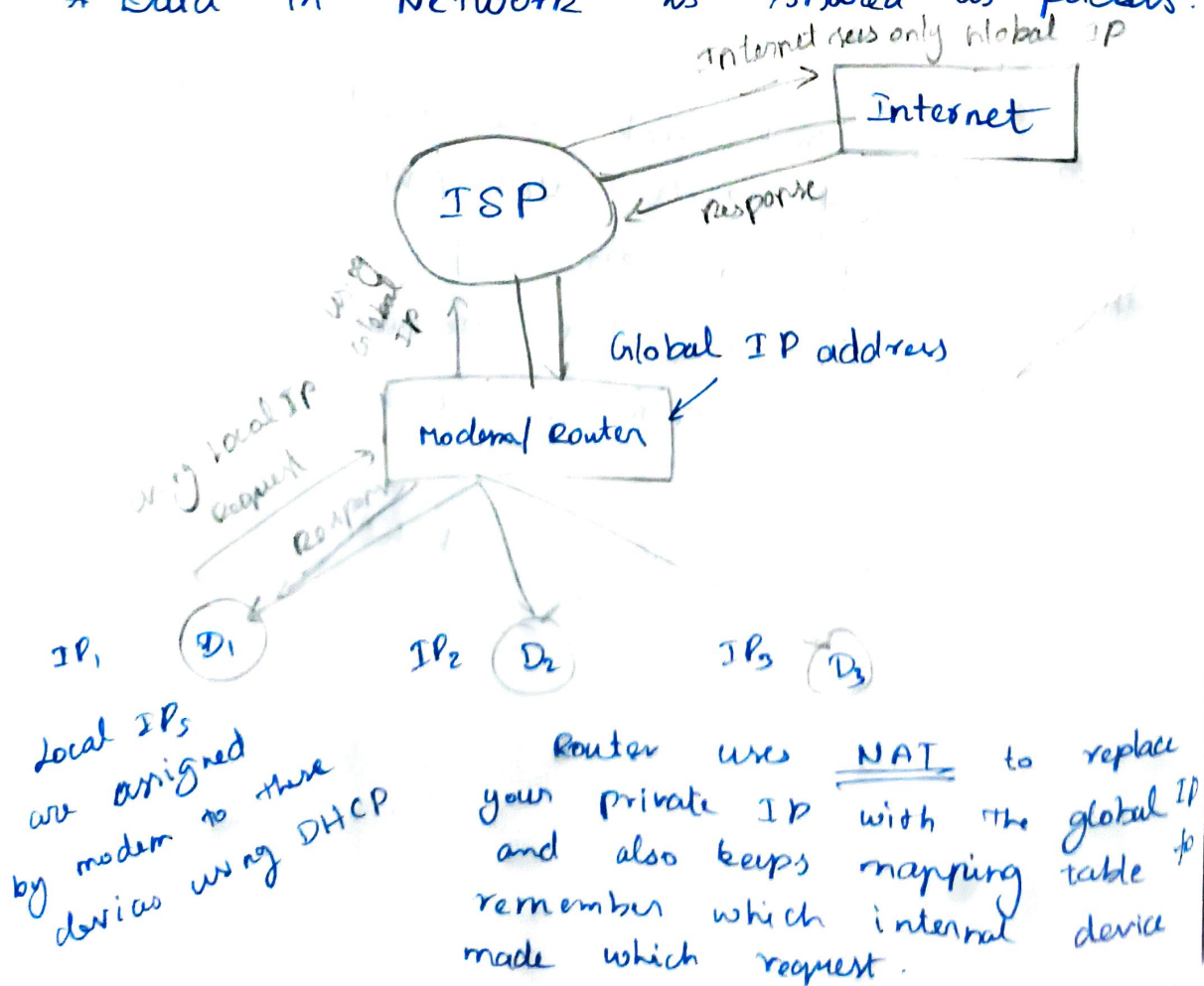
Internet Society make these rules.



Common protocols:-

TCP, HTTP, UDP, IP.

* Data in Network is shared as packets.



Port Number:

A port number is like a door inside an IP address that tells the computer which application/service should handle the data.

* An IP address identifies the device (like your house address).

* port number identifies the specific application (like the room inside the house).

NOTE:-

Port number is 16 bits long.

$$\text{total possible P.N} = 2^{16} = 65,536$$

$$\text{range} = 0 - 65,536$$

1) Well known port (0 - 1023)

Reserved for std. services (HTTP = 80, HTTPS = 443, DNS = 53 etc)

2) Registered ports (1024 - 49151)

Software/apps (MySQL = 3306, PostgreSQL = 5432)

3) Ephemeral / dynamic

Temporary ports chosen by OS (like browser connects to google)

1 Mbps = 10^6 Bits/s

1 Gbps = 10^9 Bits/s

1 kbps = 1000 bits/s

Types of Computer Network:

1) LAN (Local Area Network):

↳ network that covers a small area like home, office, school or building

- Range : few 100 meters
- Device : computers, printers, routers, switch
- Ownership : usually private
- Speed : High speed

Example:

* Wifi in home

* Office networking (PC & printers)

2) MAN (Metropolitan Area Network) :-

Network that covers city or large campus. It connects multiple LANs within a city.

Example :

- * cable TV network

3) WAN (Wide Area Network)

Network that covers a large geographical area, such as countries or entire world

ex :

- * Internet is itself bigger WAN

4) PAN (Personal Area Network)

A very small network used around a single person.

Ex :

- * Bluetooth connections (Phone $\leftrightarrow$ earbuds)

- * smart watch connected to mobile

What is a network :

* A computer network is a set of computers sharing resources located on or provided by network nodes.

* Computer use common communication protocols over digital interconnections to communicate with each other.

* These interconnections are made up of telecommunication network technologies based on wired, optical, wireless radio-frequency methods.

Host :

* A device connected to a ~~serv~~ computer network. (end device that has an IP & act as source or destination of a network)

* A host may work as a server

Servers provide sharing service such as sharing resources to its clients.

Your laptop may act as a client when it receiving any service or it can act as a server when it providing any resource.

On internet we use names such as google.com, that name is translated into an IP address using a protocol called DNS (Domain Name System).

MAC ADDRESSES: Hardware addresses burnt into a Network Interface card into a device

Communication between devices on Network:

we use protocols or languages to communicate with one another.

Such as HTTP, HTTPS, IP

Network Interface card (NIC):

NIC also called as Network interface controller is a hardware by which a device (for example a PC) connects to a network.

e.g. 3Com NIC

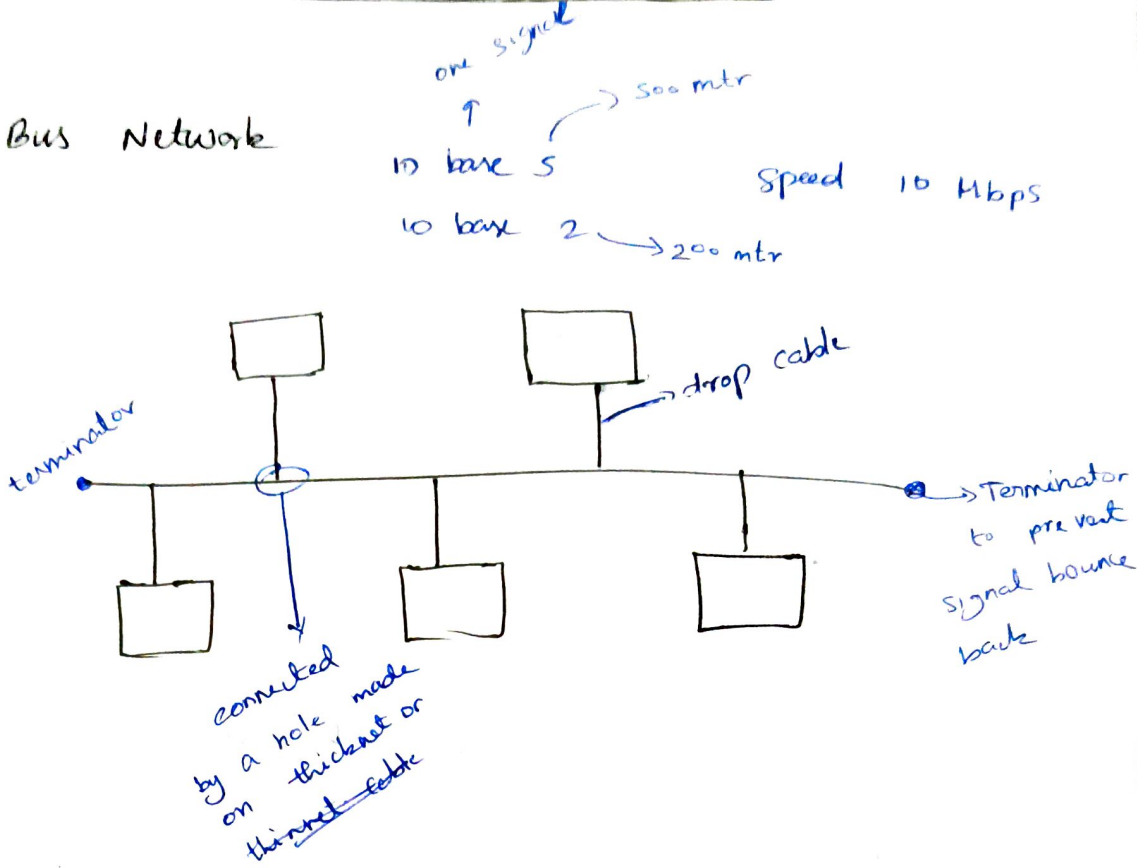
$$\left. \begin{array}{l} \rightarrow \text{RJ45 UTP} \\ \rightarrow \text{10 base 2 (Thinnet)} \\ \rightarrow \text{10 base 5 (Thicknet)} \end{array} \right\} \text{old}$$

SFP (Small form-factor pluggable) $\Rightarrow$ Fibre optic cable

when a NIC hardware manufacturer makes the card burns a MAC address into the NIC.

MAC address $\Rightarrow$ 12 hexa decimal digits (1 = 4 bits)
48 bits

Bus Network



* Thinner doesn't need drop cable.

Collision:

It happens when two devices send data at the same time on the same network medium.

unicast:

Sending data from one specific device to another specific device.

eg. whatsapp message to friend